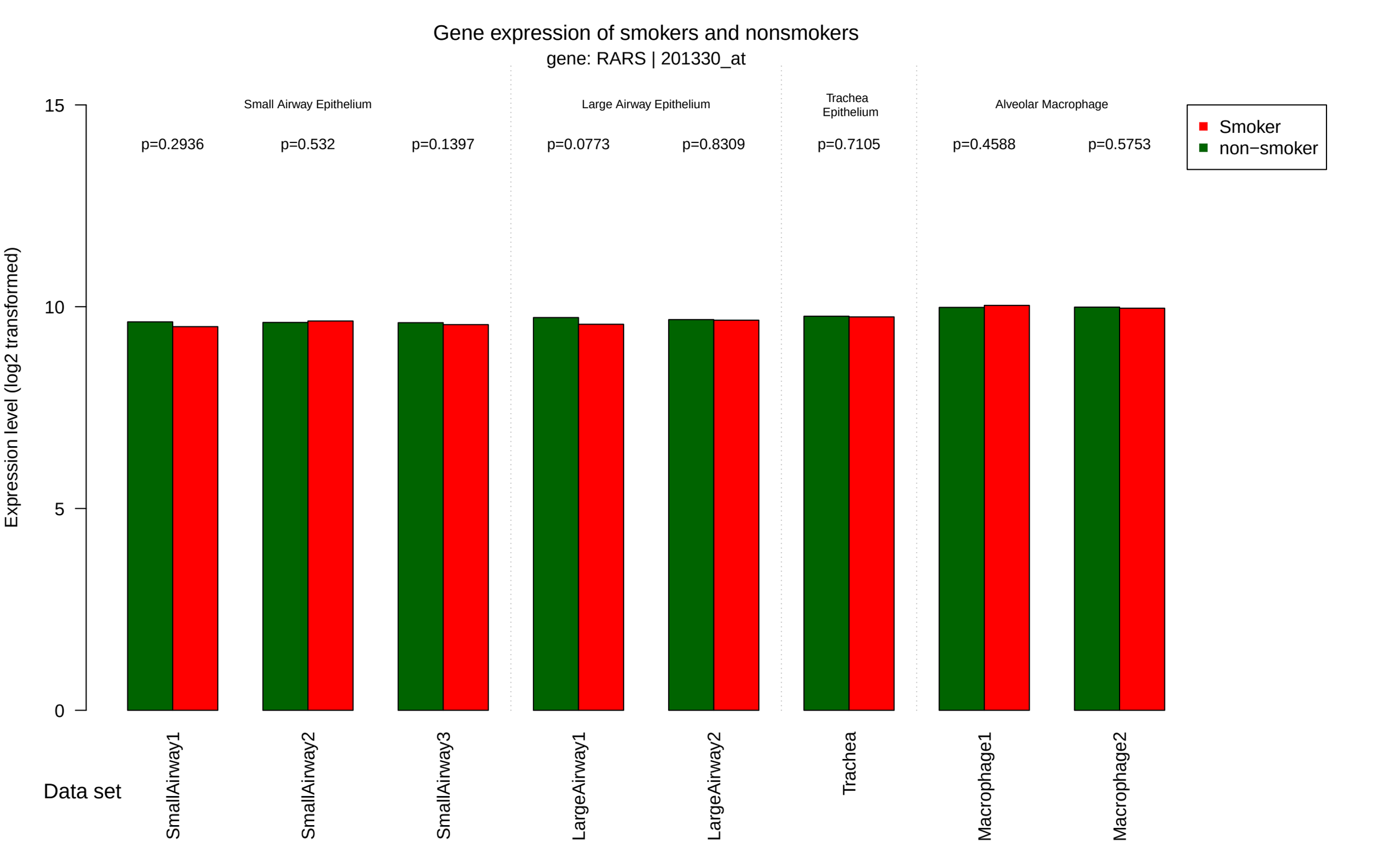
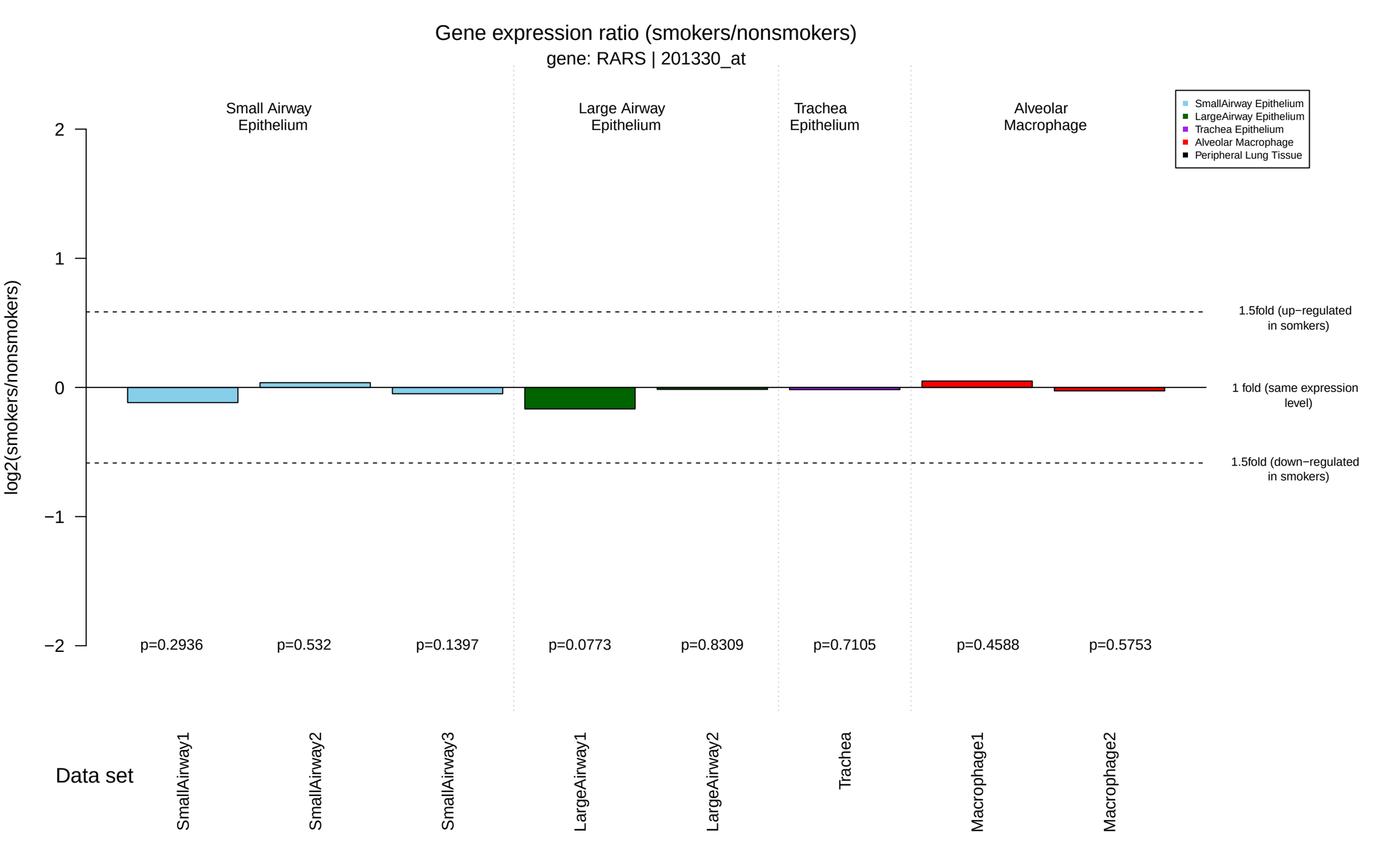
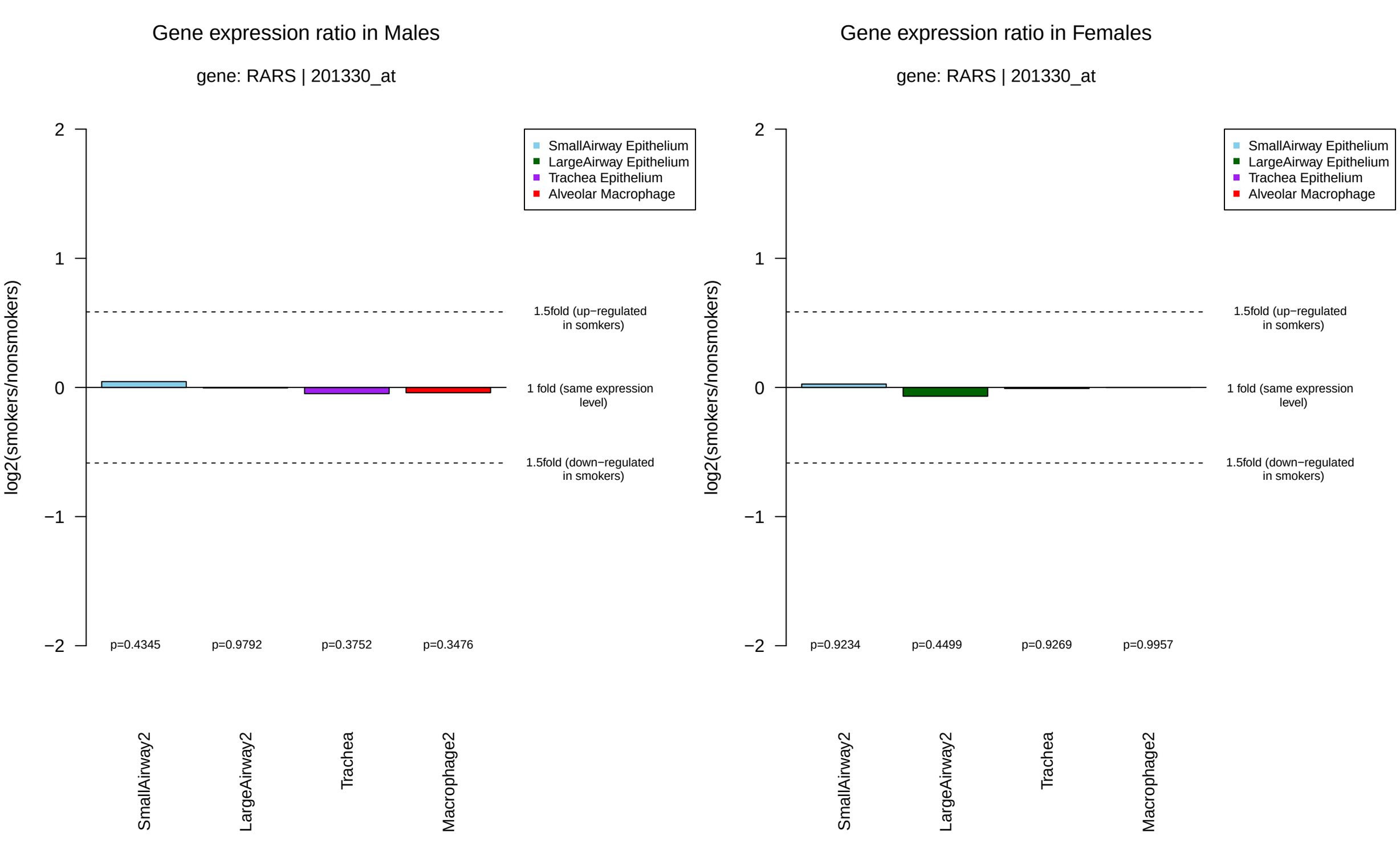
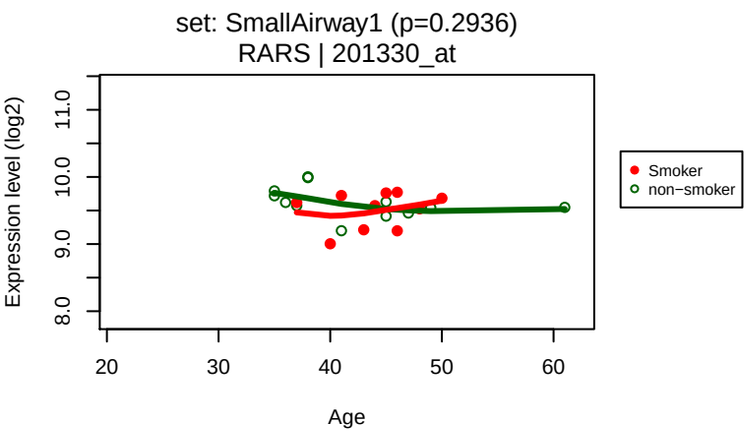
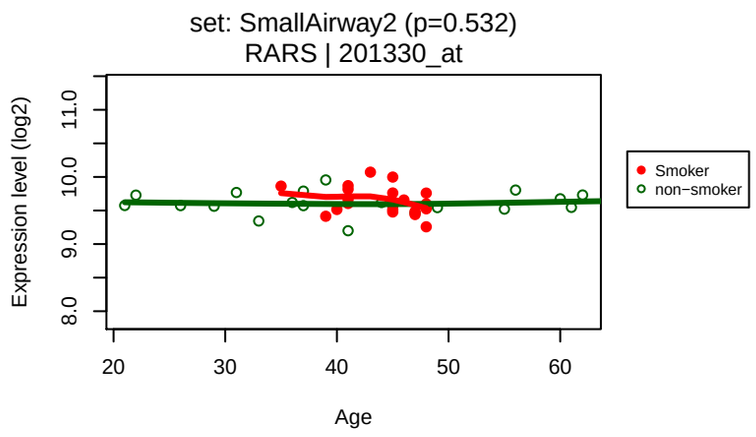
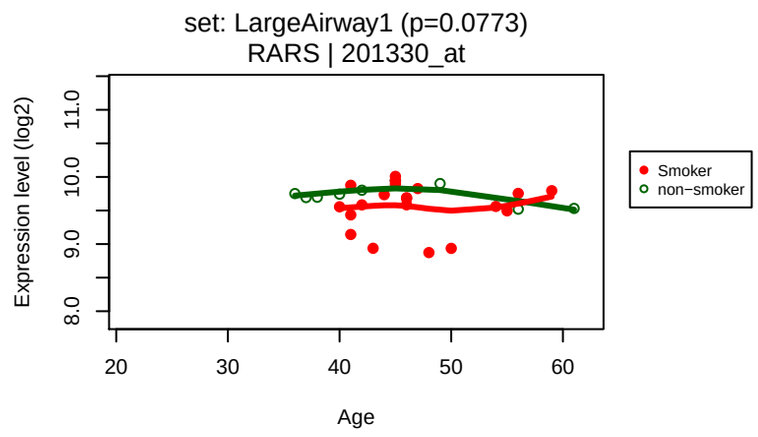

Gene expression of smokers and nonsmokers
gene: RARS | 201330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2936
p=0.532
p=0.1397
p=0.0773
p=0.8309
p=0.7105
p=0.4588
p=0.5753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

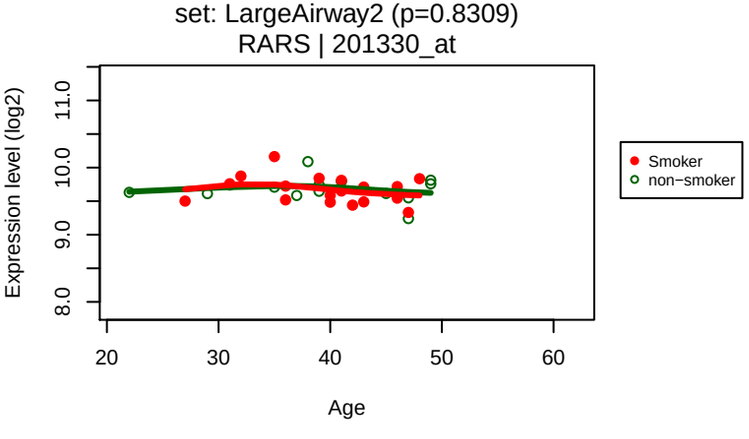

Gene expression ratio (smokers/nonsmokers)
gene: RARS | 201330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2936
p=0.532
p=0.1397
p=0.0773
p=0.8309
p=0.7105
p=0.4588
p=0.5753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

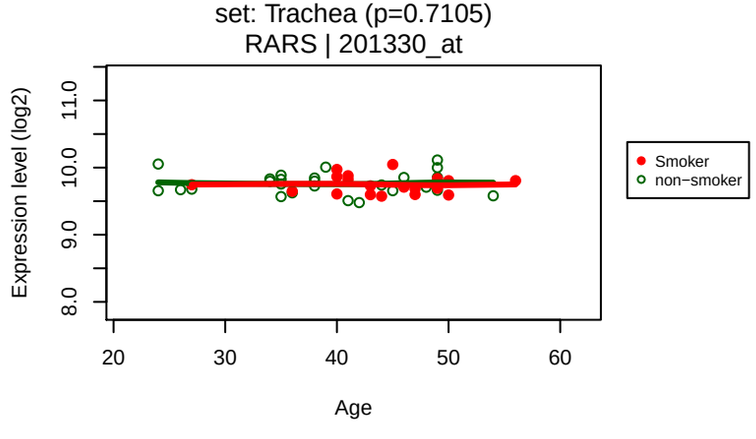

Gene expression ratio in Males
gene: RARS | 201330_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4345
p=0.9792
p=0.3752
p=0.3476
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARS | 201330_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9234
p=0.4499
p=0.9269
p=0.9957
SmallAirway2
LargeAirway2
Trachea
Macrophage2

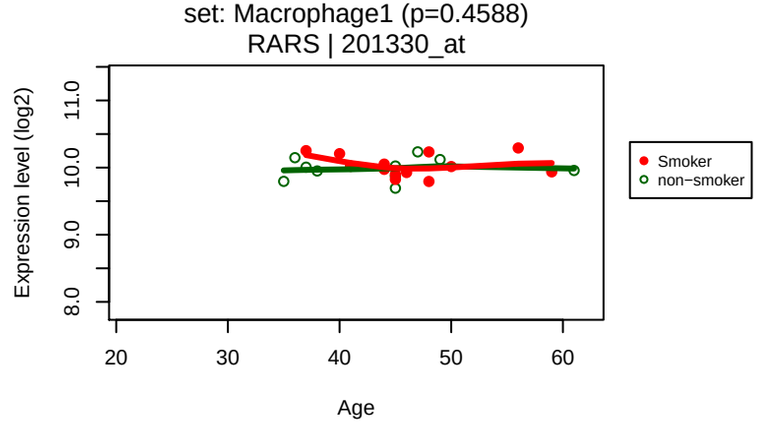

set: SmallAirway1 (p=0.2936)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker

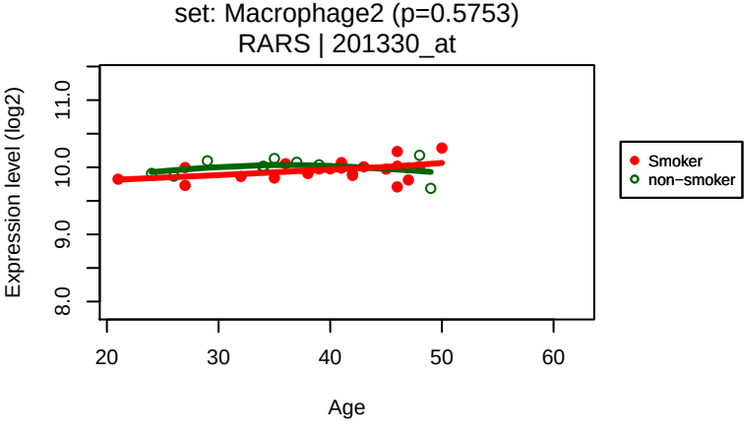

set: SmallAirway2 (p=0.532)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker

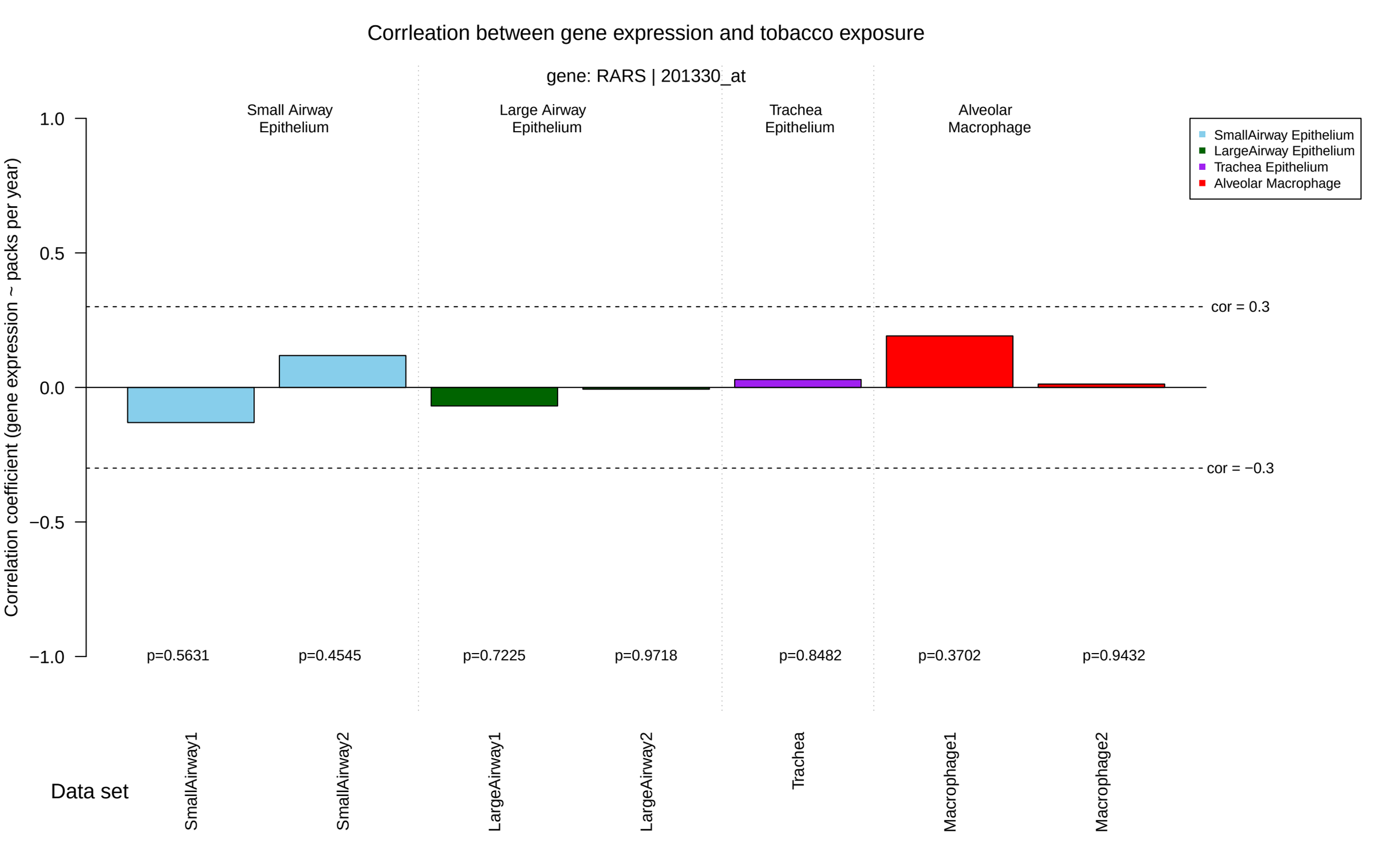

set: LargeAirway1 (p=0.0773)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8309)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7105)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4588)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5753)
RARS | 201330_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARS | 201330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5631
p=0.4545
p=0.7225
p=0.9718
p=0.8482
p=0.3702
p=0.9432
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2